



Tangents

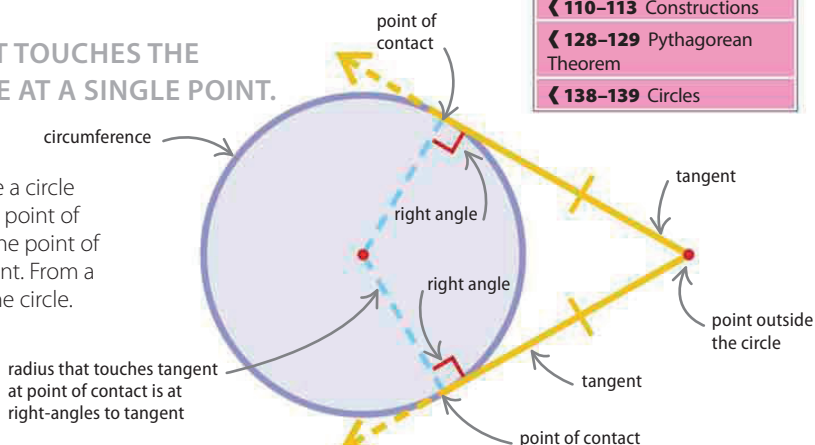
A TANGENT IS A STRAIGHT LINE THAT TOUCHES THE CIRCUMFERENCE (EDGE) OF A CIRCLE AT A SINGLE POINT.

What are tangents?

A tangent is a line that extends from a point outside a circle and touches the edge of the circle in one place, the point of contact. The line joining the centre of the circle to the point of contact is a radius, at right-angles (90°) to the tangent. From a point outside the circle there are two tangents to the circle.

► Tangent properties

The lengths of the two tangents from a point outside a circle to their points of contact are equal.



SEE ALSO

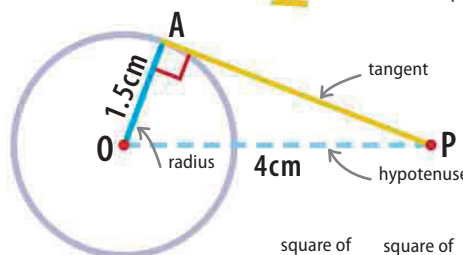
◀ 110–113 Constructions

◀ 128–129 Pythagorean Theorem

◀ 138–139 Circles

Finding the length of a tangent

A tangent is at right-angles to the radius at the point of contact, so a right triangle can be created using the radius, the tangent, and a line between them, which is the hypotenuse of the triangle. Pythagorean theorem can be used to find the length of any one of the three sides of the right triangle, if two sides are known.



◀ Find the tangent

The tangent, the radius of the circle, and the line connecting the center of the circle to point P form a right triangle.

Pythagorean theorem shows that the square of the hypotenuse (side facing the right-angle) of a right triangle is equal to the sum of the two squares of the other sides of the triangle.

Substitute the known numbers into the formula. The hypotenuse is side OP, which is 4cm, and the other known length is the radius, which is 1.5cm. The side not known is the tangent, AP.

Find the squares of the two known sides by multiplying the value of each by itself. The square of 1.5 is 2.25, and the square of 4 is 16. Leave the value of the unknown side, AP² as it is.

Rearrange the equation to isolate the unknown variable. In this example the unknown is AP², the tangent. It is isolated by subtracting 2.25 from both sides of the equation.

Carry out the subtraction on the right-hand side of the equation. The value this creates, 13.75, is the squared value of AP, which is the length of the missing side.

Find the square root of both sides of the equation to find the value of AP. The square root of AP² is just AP. Use a calculator to find the square root of 13.75.

Find the square root of the value on the right, and round the answer to a suitable number of decimal places. This is the length of the missing side.

$$\begin{aligned}
 & \text{square of one side} \quad \text{square of other side} \quad \text{square of the hypotenuse} \\
 & \mathbf{a^2 + b^2 = c^2} \\
 & \mathbf{1.5^2 + AP^2 = 4^2} \quad \text{the value of the tangent is unknown} \\
 & \mathbf{2.25 + AP^2 = 16} \quad \begin{array}{l} 1.5 \times 1.5 = 2.25 \\ 4 \times 4 = 16 \end{array} \\
 & \text{subtract 2.25 from both sides to isolate the unknown term} \quad \text{2.25 must be subtracted from both sides to isolate the unknown} \\
 & \mathbf{AP^2 = 16 - 2.25} \\
 & \text{this means AP} \times \text{AP} \quad \mathbf{AP^2 = 13.75} \quad \begin{array}{l} 16 - 2.25 = 13.75 \\ \text{this is the sign for a square root} \end{array} \\
 & \text{the square root of AP}^2 \text{ is just AP} \quad \mathbf{AP = \sqrt{13.75}} \\
 & \text{3.708... is rounded to 2 decimal places} \\
 & \mathbf{AP = 3.71cm}
 \end{aligned}$$